Understanding Traffic in and around Yellowstone National Park after the June 2022 Flooding

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Context

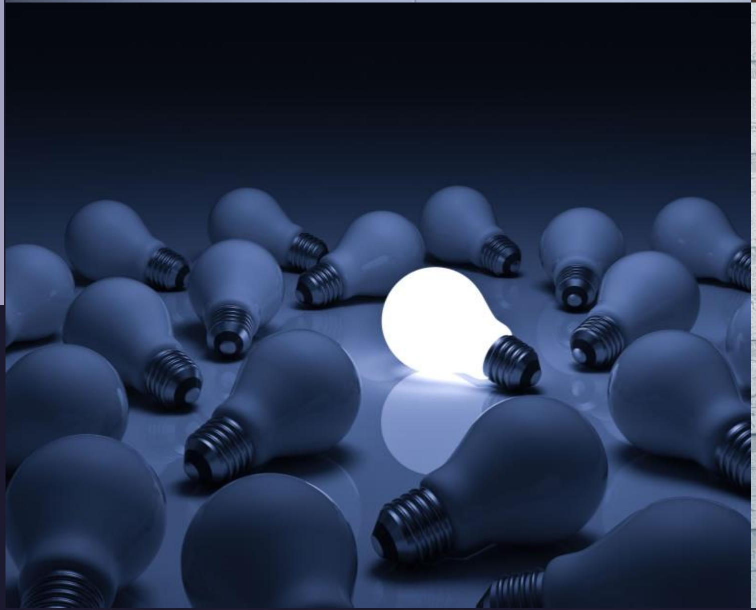
June 13th-17th 2022:

1:500-year flooding event happened in and around Yellowstone National Park (YNP) from Yellowstone River

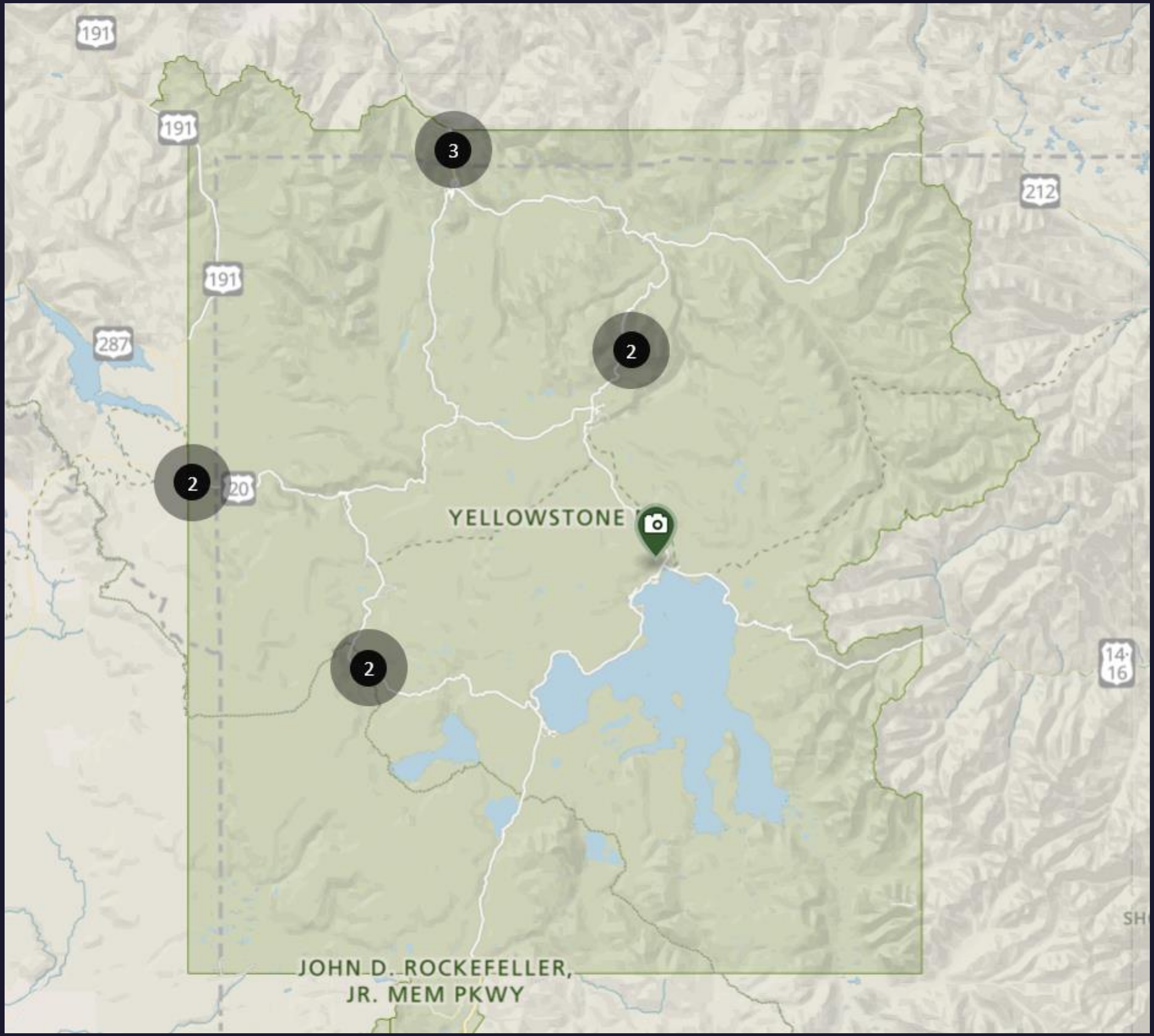
Ryan McManamay was present in Yellowstone at the time of the floods

Data collection began that same week

1) Construction
2) Roads
3) Power Outages
4) Traffic Cams



Webcam Images



https://www.nps.gov/yell/learn/photosmultimedia/webcams.htm

251 webcams

251 images – every 15 minutes

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June 18th, 2022 - Nov 17th, 2022

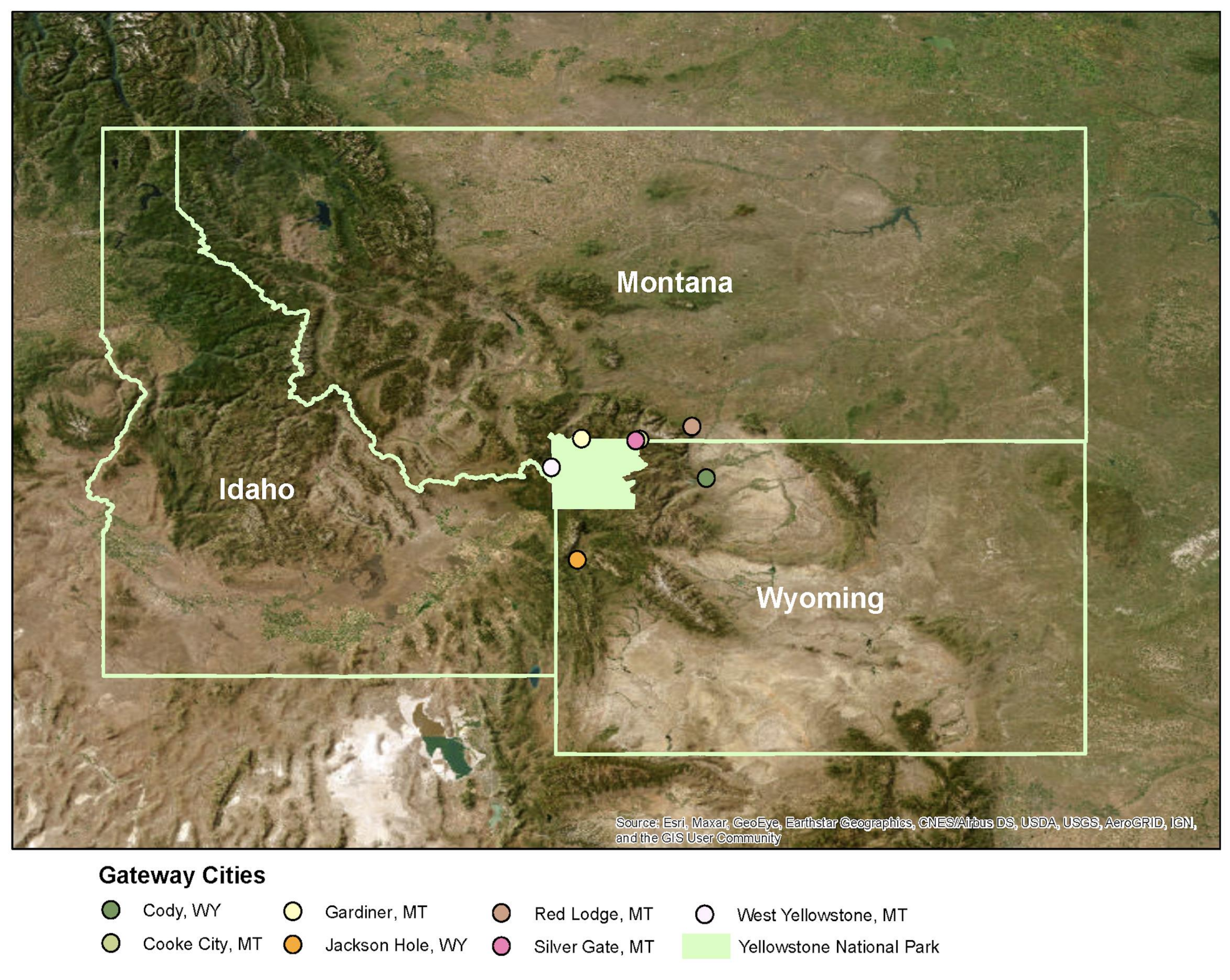
- break for the winter –

April 7th, 2023 – Present

_________(345 days)__________

~361,440 images/day

~124,696,800 total images

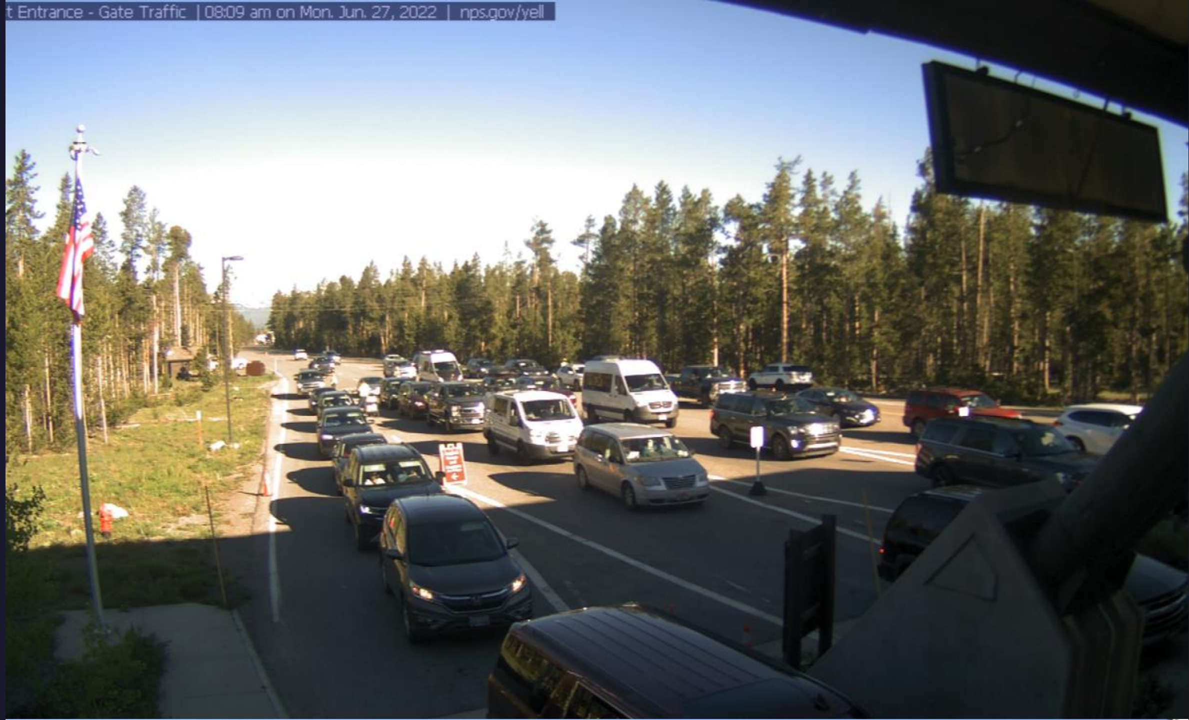


Webcam images are being collected beyond the park boundary

But if we want to understand traffic in the context of the flood, we should focus on areas in or proximate to YNP

Example of Images Collected

West Entrance Gate



June 27th, 2022
8:15am CMT

North Entrance Gate

Why is there such a difference in number of visitors?

Context for difference in camera images:

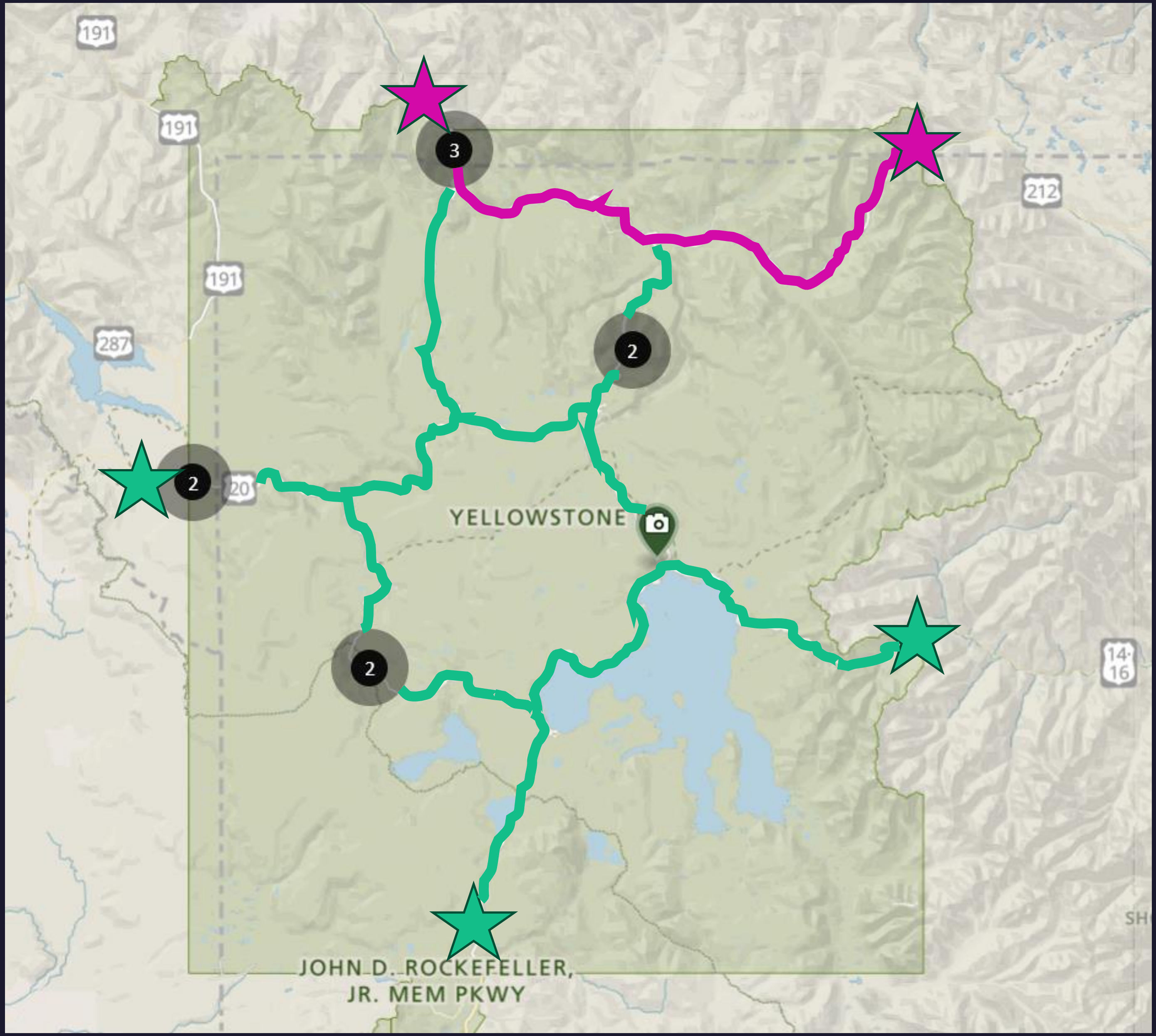
1. Flood damage mostly localized to North Yellowstone

2. Left 3/5 exits open for public use

3. But that was not exactly portrayed in the media

But with 2/5 gates closed from results of the flood, how was automobile traffic from eco-tourism directed?

Specifically, how did automobile traffic affect the other 3/5 gates?



How was automobile eco-tourism in/around YNP affected in the aftermath of the June 2022 flooding?

To answer our question, we need to quantify automobiles from our webcam images

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We are interested in the use of Mask R-CNN or Fast R-CNN to quantify the number of vehicles in or around YNP at the time of the event and over the course of a year